

Cross-lingual alignment of Contextual Word Embedding for Low-resource Indian Languages

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under the guidance of

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CERTIFICATE

*This is to certify that the work contained in this thesis entitled “**Cross-lingual alignment of Contextual Word Embedding for Low-resource Indian Languages**” is a bonafide work of **Animesh Ranjan (Roll No. 194101005)**, carried out in the M.Tech programme, Indian Institute of Technology Guwahati under my supervision and that it has not been submitted elsewhere for a degree.*

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Abstract

Cross-lingual Contextual Word Embedding (CLWEs) are representation of words in a vector form which allows us to retain its contextual knowledge while generating the embedding. Existing supervised models for NLP task generally rely on large annotated corpora which available is not for all languages, however collecting of small annotated corpus is feasible but due to data sparsity it lacks in the performance. Cross-lingual representation facilitate us to provide language processing models which are suitable for low-resource languages as it provides us better performance in closely related languages. In Indian low resource languages cross lingual alignment of words can significantly increase the performance as mostly of them share similar vocabulary and semantics. In this paper we proposed a Fine-tune Model for existing BERT model with help of parallel corpora with objective of explicit alignment of Word Embedding for better Cross-Lingual Transfer Learning.

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Chapter 1

Introduction

1.1 Introduction to Word Embedding

Word embedding are used to represent words as vector, that can be used in solving of many NLP problems. These Embedding are distributed representations of text in an n-dimensional space. Loosely speaking, word embedding are a form of vector representation of a particular word, which is capable in capturing of the context of a word in document along with its syntactic and semantic similarity with other words. These word embedding can be broadly classify into two types: Frequency Based and Prediction Based Embedding

1.1.1 Frequency Based Embedding

One of the basic ways to represent a word in numerical form is Count Vector as discussed in [13]. Here the idea is simple, author creates a vector that has the same number of dimensions as the number of unique words in the Corpora used. Each Unique word will be assigned a unique dimension and it represented by 1 in that particular dimension and with 0s everywhere else. As result of this we will be getting a huge and sparse vector that will contain no relation information also no semantic relationship information.

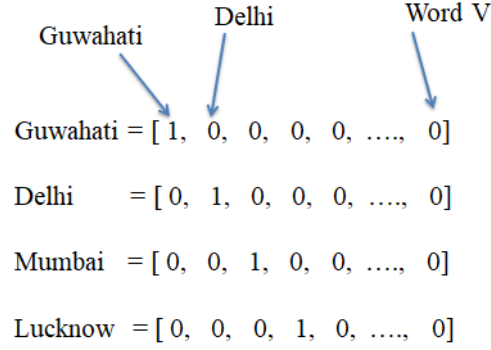


Fig. 1.1 Vector Representation of Words

1.1.2 Prediction Based Embedding

These method use prediction based algorithm in order to determine the probability that a word vector is similar with other words vectors of the words present in the corpora. One of the basic such implementation is Word2vec [7], Word2vec is a combination of two techniques - CBOW(Continuous Bag of Words) [6] and Skip-gram model [6].

The wordking principle of CBOW is this method learns an embedding of the word by predicting the current word embedding based on the context of the word and Skip-gram model used to learns surrounding of the words while it have the context of word. High Quality embedding can be learned by using this technique and also we can get more features in our representation due to larger dimensional.

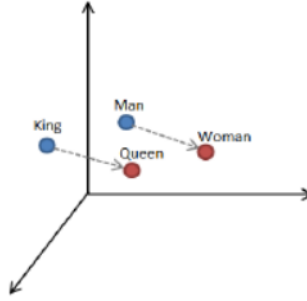


Fig. 1.2 Word2vec : Word Embedding [7]

1.2 Contextual Word Embedding

The need of word embedding is to provide an exact meaning to the word, but the meaning of the word changes based on the context it is been used. Consider the Example below:

I **left** my phone at home.

My house is at **left** side of the road.

Here we can see the meaning of the word **left** changes with respect to the context it has been used. So it is important to use a method that captures the word meaning while the context is changing, also it should be capable of retaining the contextual information by learning sequence-level semantics considering sequence of all the words in the document, some of the methods are ELMo ,BERT which are discussed further in the review.

1.3 Cross-Lingual Word Embedding

Cross-Lingual word embeddings (CLWEs) are n-dimensional vector space representation of word embedding that is used for multiple languages. This uses the additional insight that many words in all languages refer to common concept. CLWEs are useful because of mainly two reasons: firstly, they provide us ease to compare meaning of words across different languages, secondly CLWEs also provide concept of model transfer between languages so is we can take its advantage in Low-Resource language.

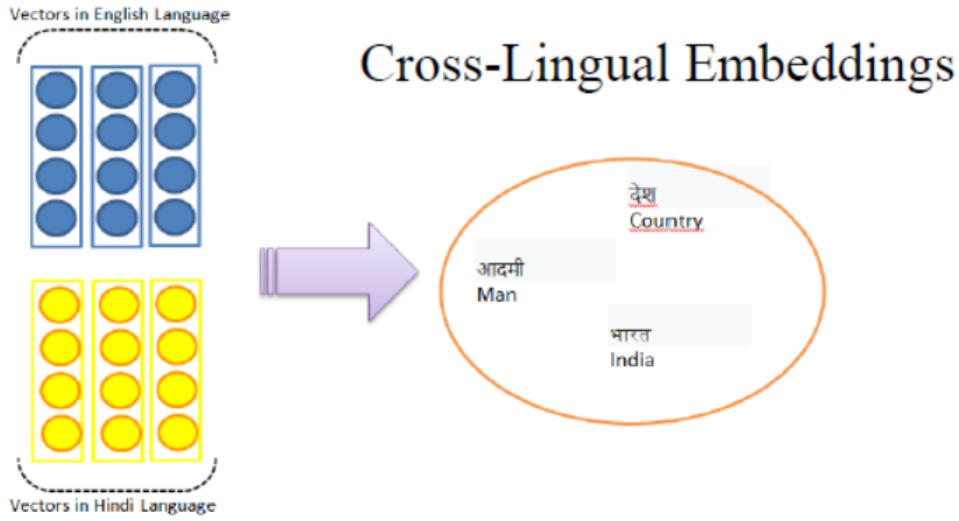


Fig. 1.3 Ex: CLWE for English and Hindi

1.4 Problem Statement

Many supervised solution exists for Named Entity Recognition but they require large annotated corpora. So, for low-resource languages Cross-Lingual learning is suitable, briefly discussed in Chapter 2. Cross Lingual alignment of words in universal space has a major impact on Contextual Word Embedding across languages. Our Motivation here is to align Embedding of different language in share space to increase the efficiency of the model.

1.5 Organization of The Report

In this report, we have gone through several research papers, that have been based on this domain, i.e. in general what is word embedding and how to generate contextual-level word embedding. In chapter 1 we briefly discussed about what is word embedding, contextual-level word embedding and its importance. In upcoming Chapter 2 we will discuss some major breakthrough in this domain. In Chapter 3 we will discuss the proposed scheme for the problem statement. In Chapter 4, we will discuss our experiment and results. In chapter 5 we will conclude our work and discuss the future scope.

Chapter 2

Related Work

Significant amount of work has been done in this domain. The initial approaches were based on representing the words in a corpora into a n-dimension vector space such that each word has its unique vector representation as developing a meaningful representation of words has been one of the primary goal of NLP. The foundation task has been during 2010s and later followed by significant improvement in capturing word sense and similarity in different context.

In upcoming points we are going to discuss about the literature review of Contextual Word Embedding evolution i.e. from initial stages of CLWEs to more advanced and modern methods to generate better CLWEs.

2.1 ELMo: Deep contextualized word representations [10]

In this paper for determining the word embedding in place of using traditional fixed embedding for each word, what ELMo does is it looks at the entire sentence first then assign embedding to each word, so that it take care for better contextual level information of the word i.e. in that particular sentence what is contextual meaning of the word. For that it uses a bi-direction LSTM (Long-Short Term Memory) that are a type of recurrent neural network which are capable of learning order dependence for sequence prediction problem.

These ELMo models both aspect i.e. complex characteristics of word use: syntax and semantics and how these uses vary across linguistic contexts. These word embeddings are learned from the internal state of bi-LSTM, which is pre-trained on a Large text corpus.

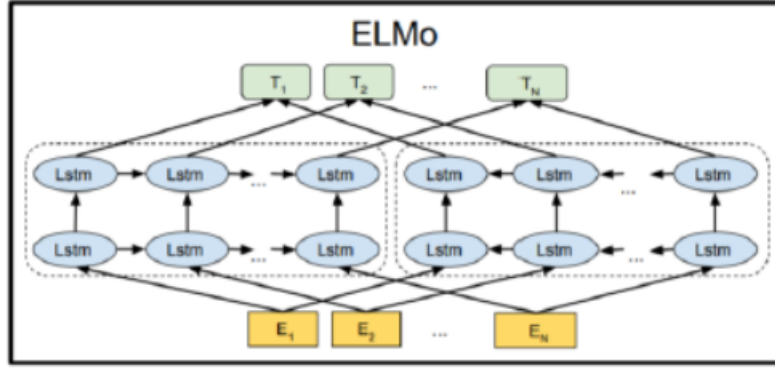


Fig. 2.1 ELMo Architecture [10]

2.2 Bert: Pre-training of deep bidirectional transformers for language understanding [1]

In [1] the author developed a non recurrent model based on transformer model [14] as they use self attention which will help in alignment for translation in seq2seq manner to capture the relationship between the words in a sentence of the corpora. BERT is basically a multi-layer transformers encoder. The paper here consist of two models BERT base for this they used 12 layer transformer block having 12 attention head and 110 million parameters, and second model was BERT Large having 24 layers of transformer block having 16 attention head and 340 parameters, the model was then pre-trained on 2 tasks first was Masked Language Modeling, in this 15% of random tokens are masked and the objective was to predict only those masked tokens, second was Next Sentence Prediction here the task was a pair of sentence is given the model has to just predict if second sentence follows first one or not. The tokenizer used in this paper is WordPiece tokenization [12] also this model is good for many NLP task because of ease in fine-tuning as per requirement.

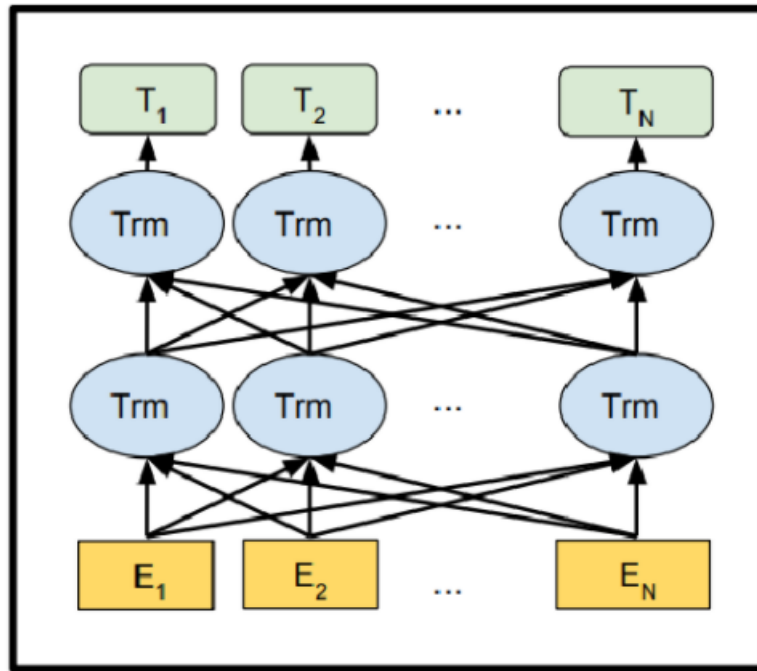


Fig. 2.2 BERT Architecture [1]

2.3 *Improving NER Tagging Performance in Low-Resource Languages via Multilingual Learning* [8]

In [8] the author developed a model to create word embedding specifically for low-resource Indian languages, firstly for the model they have to create an in-house dataset by combining corpus of different Indian languages i.e. Hindi, Marathi, Bengali, Tamil and Malayalam to make a big corpora, here Hindi Corpus is relatively big and has annotated data, the problem of languages having different script are tackled by mapping them to Devanagari Script. For training of the model the author uses bi-LSTM in order to learn embedding of word from left-to-right and right-to-left sequence. Three training strategies were used first was simple Monolingual training where the model is trained on primary language only, second was Share Subword features where subword information between target corpus and training corpus were shared and last was All share i.e. a multilingual model which was trained language independent way. The architecture consists of a word feature extractor which has a lookup table which is initialized with pre-trained word embeddings then a subword feature extractor which consists of CNN layers which take character sequence as input and given subword representation as output, followed by a Disambiguation layer which is bi-LSTM then finally an output softmax layer.

2.4 IndicNLP Suite: Monolingual Corpora, Evaluation

Benchmarks and Pre-trained Multilingual Language Models for Indian Languages[3]

In this paper authors developed NLP resources specifically for 11 major Indian languages, which consist of

- IndicCorp: A Large scale sentence-level monolingual corpora.
- IndicGLUE: Multilingual NLU benchmark like GLUE[15].

- IndicFT: A FastText word embeddings for each languages using IndicCorp.
- IndicBERT: A multilingual NLU model which is trained on IndicCorp and evaluated on IndicGLUE

, To create this model ALBERT [1] is used as base model, for pre-training firstly they use a sentence piece tokenizer then this tokenized corpora is trained on multilingual ALBERT [1] using standard masked language model technique and for fine-tuning it has been done independently for every task. This IndicNLPSuite can be used for various NLP tasks related to Indian Languages.

Chapter 3

Proposed Scheme

Our problem statement it to improve the performance of CLWE of Low-resource languages for Named Entity Recognition and other NLP tasks, Cross Lingual Word Alignment is prominent technique to improve performance of existing SOTA models. Existing SOTA models for CLWE such as ABERT [5] requires large monolingual corpora, and other efficient models for CLWE is not purely bidirectional. BERT pre-trained model is most widely used in many down-stream NLP tasks, but bert has no explicit alignment of these word embedding together in the shared space. So we proposed a Multilingual Alignment method using parallel corpora with word-based alignment objective. In this parallel corpora of different language set is taken i.e. source and target and two different attention masks are created by aligning the corpora firstly in source-to-target manner and secondly by target-to-source then distance between those mask are minimized which gives final attention mask which is used to perform fine tuning of BERT to give better aligned contextual word embedding, the evaluation is performed by NER performance of our model on WikiAnnNER dataset for 11 Indian Languages.

3.1 Cross-Lingual Alignment

Here our objective was to align Cross-Lingual Word Embedding, for that we used

3.1.1 Word Alignment

Here our objective is to create attention mask with help of parallel language pair corpora that encourages alignment of word embedding by leveraging attention mechanism of the Transformer Model, here our motivation was to increase the consistency between the source-to-target and target-to-source translation as is paper [4] we here created two different attention mask one for source-to-target language pair and other for target-to-source language pair, by this be obtained two attention matrices as described in 3.1

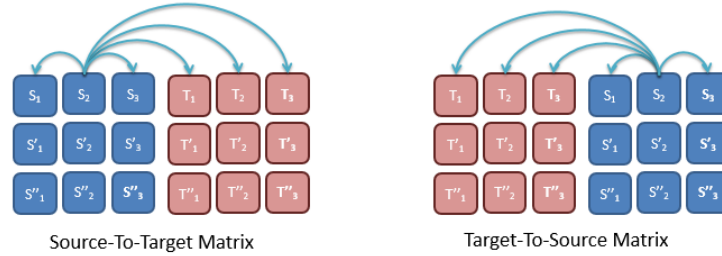


Fig. 3.1 Source-To-Target and Target-To-Source Attention Matrix

Now to align source and target words in both direction, our objective is to minimize the distance between these two attention matrices, for that our aim is to maximize the trace of two attention matrix, when score of these two are identical then final score was bounded by $\min(\text{source}, \text{target})$, as transformer generate multiple attention head we calculated average of all attention head.

3.1.2 Masked Language Modeling

Here our objective was to take sentence pair from parallel corpora and concatenate them, after that optimize the prediction of random masked word token in the concatenated corpora we took 15% of random token and replace them by [MASK]. In this firstly we tokenize our concatenated corpora like regular BertMaskedLM, then created label tensors to calculate loss, here we used previously calculated ID's and attention mask in word alignment the we added the this on top of classification layer.

Chapter 4

Experiment and Results

4.1 prerequisite

4.1.1 Pre-Trained BERT Embedding[1]

We used pre-trained BERT-Base Uncased for our model which is a stack of encoders of transformers, in BERT-Base it has 12 layers of encoder stacked together with hidden size of 768, which is pre-trained on 102 Languages, on the top of which our fine-tuning layer will be placed.

4.2 Dataset

4.2.1 Parallel Corpora

PMIndia Dataset[2]

PMIndia Dataset is a publicly available parallel corpora for Indian languages, it consist of pair of 13 major Indian languages namely Hindi(hi), Bengali(bn), Gujarati(gu), Marathi(mr), Telugu(te), Tamil(ta), Odia(or), Malayalam(ml), Punjabi(pu), Kannada(kn), Manipuri(mni), Assamese(as) and Urdu(ur) all of them paired with English, where each language pair has 56000 sentences in it.

4.2.2 NER Corpora

WikiAnnNER[9]

To check the performance of our model for Indian Languages we used Named Entity Recognition for that we have used WikiAnnNER dataset from which we have only considered the Indian Languages i.e. consist of 11 Indian Languages namely Hindi(hi), Bengali(bn), Gujarati(gu), Marathi(mr), Telugu(te), Tamil(ta), Odia(or), Malayalam(ml), Punjabi(pu), Kannada(kn) and Assamese(as) consisting of 787k examples.

WikiAnnNER	
Language	Number of Example
pa	9,462
hi	69,431
bn	109,508
or	8,687
as	6,295
gu	39,708
mr	108,579
kn	28,854
te	81,627
ml	138,888
ta	186,423
Total	787,462

Table 4.1 WikiANN Dataset Statistics

4.3 Fine-Tuning of Model

4.3.1 Tokenization & Input Formatting

To feed our corpora to BERT it is split into tokens, and these tokens are mapped to their index in tokenizer vocabulary, for that we take our initial intuition and created 2 different inputs i.e. from source-to-target and target-to-source and these sentences in the corpora are tokenized using BERT Tokenizer, for that we used `tokenize.encode_plus` function, fol-

lowed by formatting as adding special token in starting and end of each sentence created, Padding and truncating all sentences to a constant length i.e. maximum length sentence in the corpora, then different attention mask are created based on the arrangement of the sentences.

4.3.2 Attention Matrix Alignment

The attention mask created for both source-to-target and target-to-source are aligned by increasing consistency between them, the two attention masks are provided as input to Transformer model, which gives us the attention matrix, now we minimize the distance between these two matrix by maximizing the trace of two attention matrices by 4.1

$$F_{mat}(s, t) = 1 - \sum_{n=1}^N tr(A_{t \rightarrow s}^n {}^T A_{s \rightarrow t}^n) \quad (4.1)$$

4.3.3 Fine-Tuning with MLM

Here in Masked Language Modeling in place of monolingual setting we used our source-to-target concatenated corpora of parallel language pair sentences, with randomly masked tokens which includes of masking of 15% of the tokens in paired parallel language sentences.

$$\ell_{MLM}(s, t) = -E_{x \sim [1|n]} \log P(n_i | n_{i'}) \quad (4.2)$$

Here in 4.2 s is source sentence in Indian language, t in target sentence in English language, n represent concatenation of source and target sentences where n_i are masked token and rest all are unmasked. Below is training and validation loss of the model in 4.1. We get the motivation for this by Translation Language Modeling by [4]

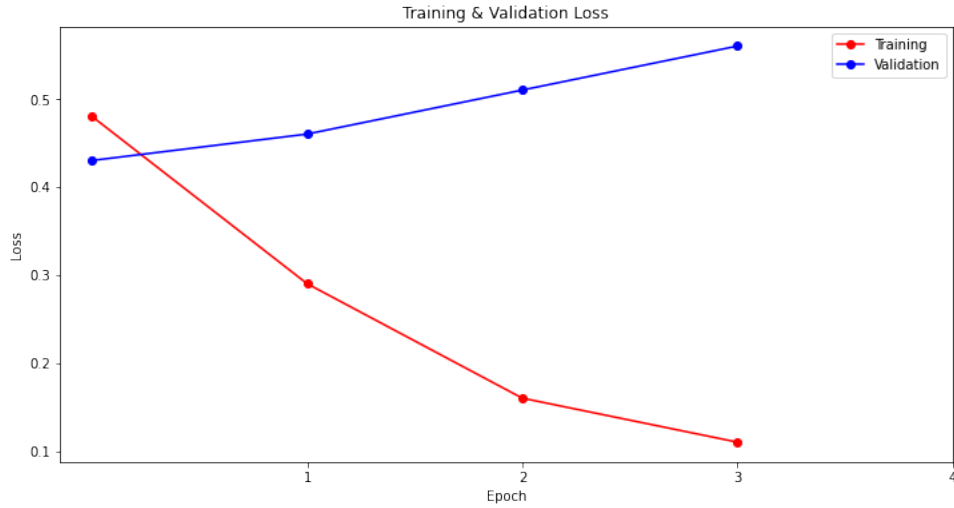


Fig. 4.1 Training and Validation Loss

4.4 Comparative Evaluation

For Evaluation of performance of our model we used WikiAnnNER[9] Dataset, mainly considering 11 Indian languages from it which consist of total 787,462 token. We did the experiment for each language individually to visualize the performance of every language for that we used *NERModel* from *simpletransformers library*[11] with learning rate $1e-4$ and calculated F1 score of our model, for few Indian languages our model performs better while for few BERT model outperform our model, as shown in 4.2, here we can visualize that over model relatively good for Low-resource language i.e. Odia(or) but for some languages such as Hindi(hi) and Assamese(as) BERT model outperformed, out model has also shown a slight increase in overall performance for all Indian Languages combined.

F1-Score		
Language	BERT	Proposed Model
pa	0.5103	0.5014
hi	0.8656	0.8105
bn	0.9181	0.9036
or	0.2105	0.4012
as	0.9231	0.8806
gu	0.6804	0.6614
mr	0.9127	0.8753
kn	0.5972	0.6315
te	0.8431	0.8512
ml	0.8268	0.8164
ta	0.7990	0.8019
Average	0.7351	0.7395

Table 4.2 F1-Score Comparision

Chapter 5

Conclusion and Future Work

In this paper we demonstrate the advantage of explicit alignment of word embedding in Cross-Lingual transfer learning for downstream NPL tasks for Low Resource Languages. As speaking of future work, we can use any Alignment tool to generate alignment score of source and target language pair, accordingly align the word embedding of words with help of alignment score.

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